Model comparison and diagnosis

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Data setup: main simulation

Simulated data dimensions:

Dimension	Value
Participants (P)	20
EEG electrodes (C)	61
Conditions (K) (noise; noise + signal)	2
Time points (T)	51 (0–500 ms, 100 Hz)
Trials per participant per condition (R)	20

Raw data: $P \cdot C \cdot K \cdot T \cdot R = 2{,}488{,}800$ rows.

Aggregated data (grouped by ppt, electrode, condition, time): $P \cdot C \cdot K \cdot T = 124{,}440$ rows.

Ground truth: noise model

Each trial has additive noise:

$$\text{noise} \;=\; \text{temporal noise} \;+\; \text{spatial noise}$$

Temporal 1/f noise (Yeung et al., 2004): 50 sinusoids with EEG-like power spectrum, $\mathrm{Var} = \sigma^2_{\text{out}} = 0.5$, drawn independently for each (trial, electrode).

Ground truth: signal model

$$\text{signal} \;=\; \underbrace{\text{peak amplitude}}_{A = 1.5\,\mu\text{V}} \cdot \underbrace{\text{temporal shape}}_{g(z)} \cdot \underbrace{\text{spatial weight}}_{w(c)}$$

Truncated Gaussian $g(z)$, with $z = (t - \tau_p)/\sigma_t$ and $\sigma_t = 0.060\,\text{s}$, $\tau_p \approx 0.250\,\text{s}$:

$$g(z) = \frac{\phi(z) - \phi(1.5)}{\phi(0) - \phi(1.5)}, \quad |z| \le 1.5; \quad g(z) = 0 \text{ else.}$$

ϕ = standard Gaussian density. $g(0) = 1$, $g(\pm 1.5) = 0$; window $\approx [0.160, 0.342]\,\text{s}$.

Spatial Gaussian RBF (unnormalised Gaussian, peak = 1 at CZ):

$$w(c) = \exp\!\Big(-\frac{d^2(c, \text{CZ})}{2\,\sigma^2_{\text{space}}}\Big), \quad \sigma_{\text{space}} = 0.5.$$

Spatial ground truth: active electrodes

The current simulation uses a hard-threshold spatial Gaussian.

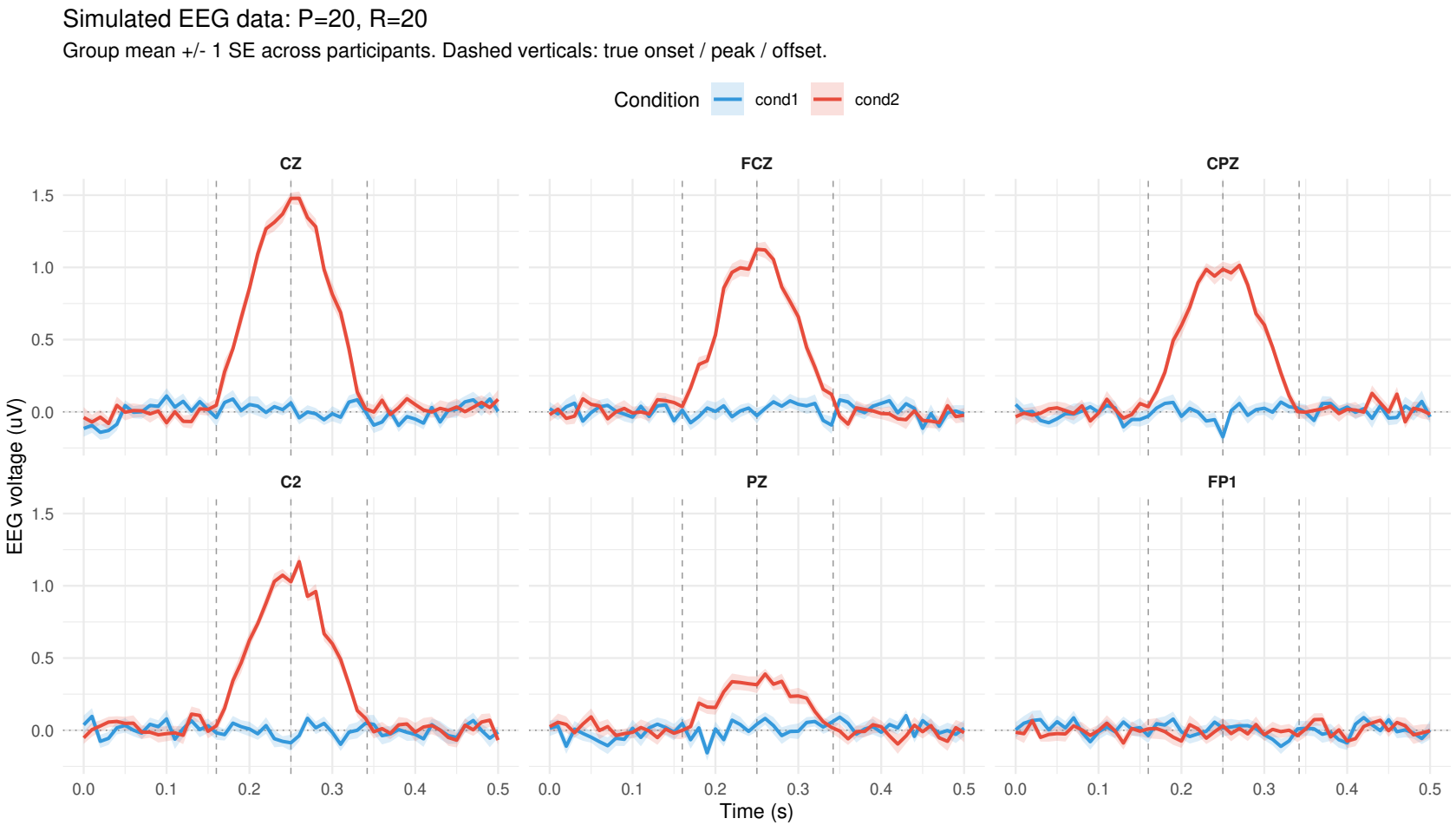
$$d(c, \mathrm{CZ}) = \|p_c - p_{\mathrm{CZ}}\|_2$$

$$w(c) = \begin{cases} \exp\left(-\dfrac{d^2(c, \mathrm{CZ})}{2\sigma_{\mathrm{space}}^2}\right), & d(c, \mathrm{CZ}) \le \sigma_{\mathrm{space}}, \\ 0, & d(c, \mathrm{CZ}) > \sigma_{\mathrm{space}}. \end{cases}$$

electrode c is active iff $d(c, \mathrm{CZ}) \le \sigma_{\mathrm{space}}$.

This is the ground-truth active-electrode list used for Spatial F1, Spatio-temporal F1, FP, and FN metrics.

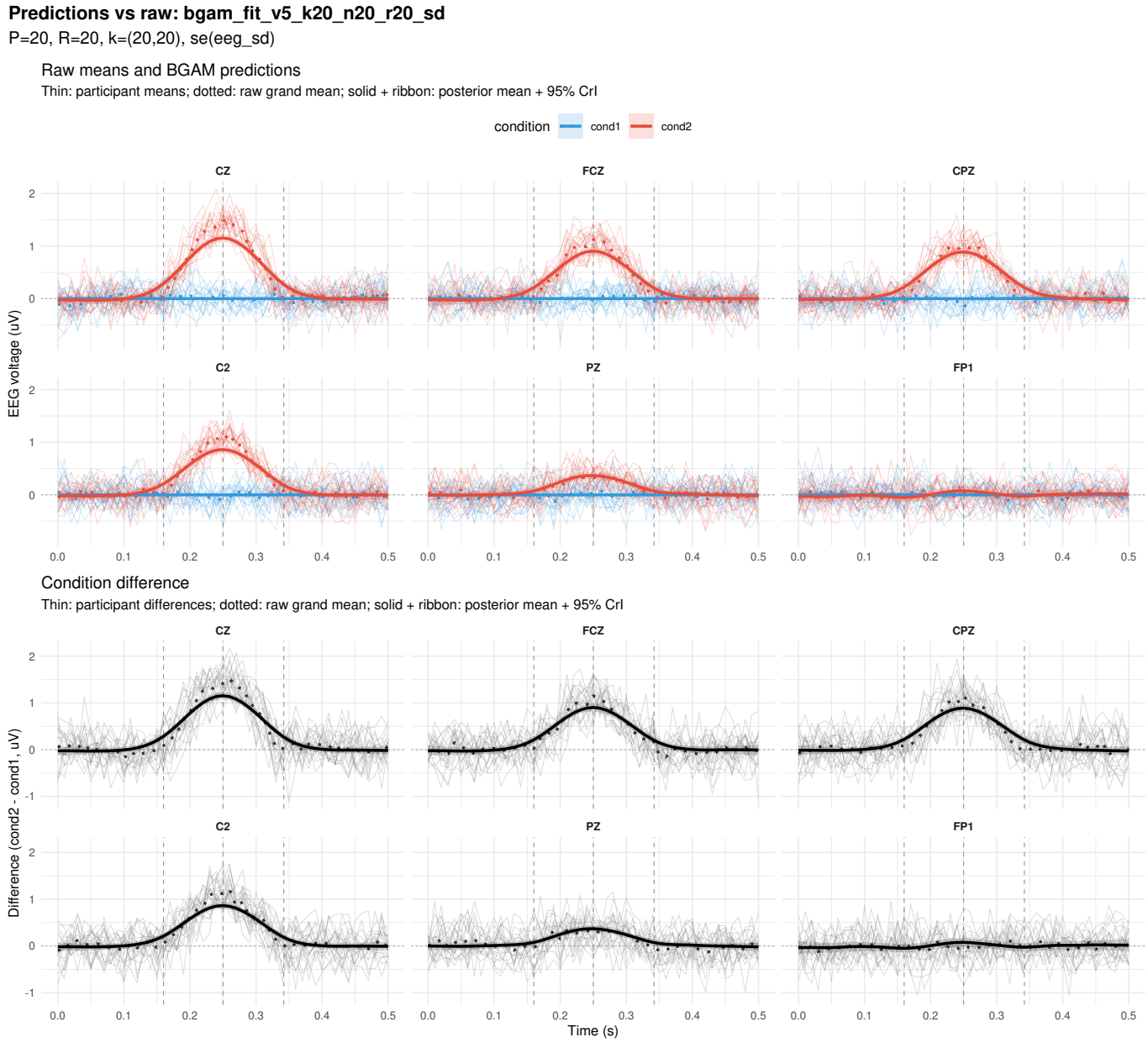
Simulated EEG data ($P = 20$, $R = 20$)



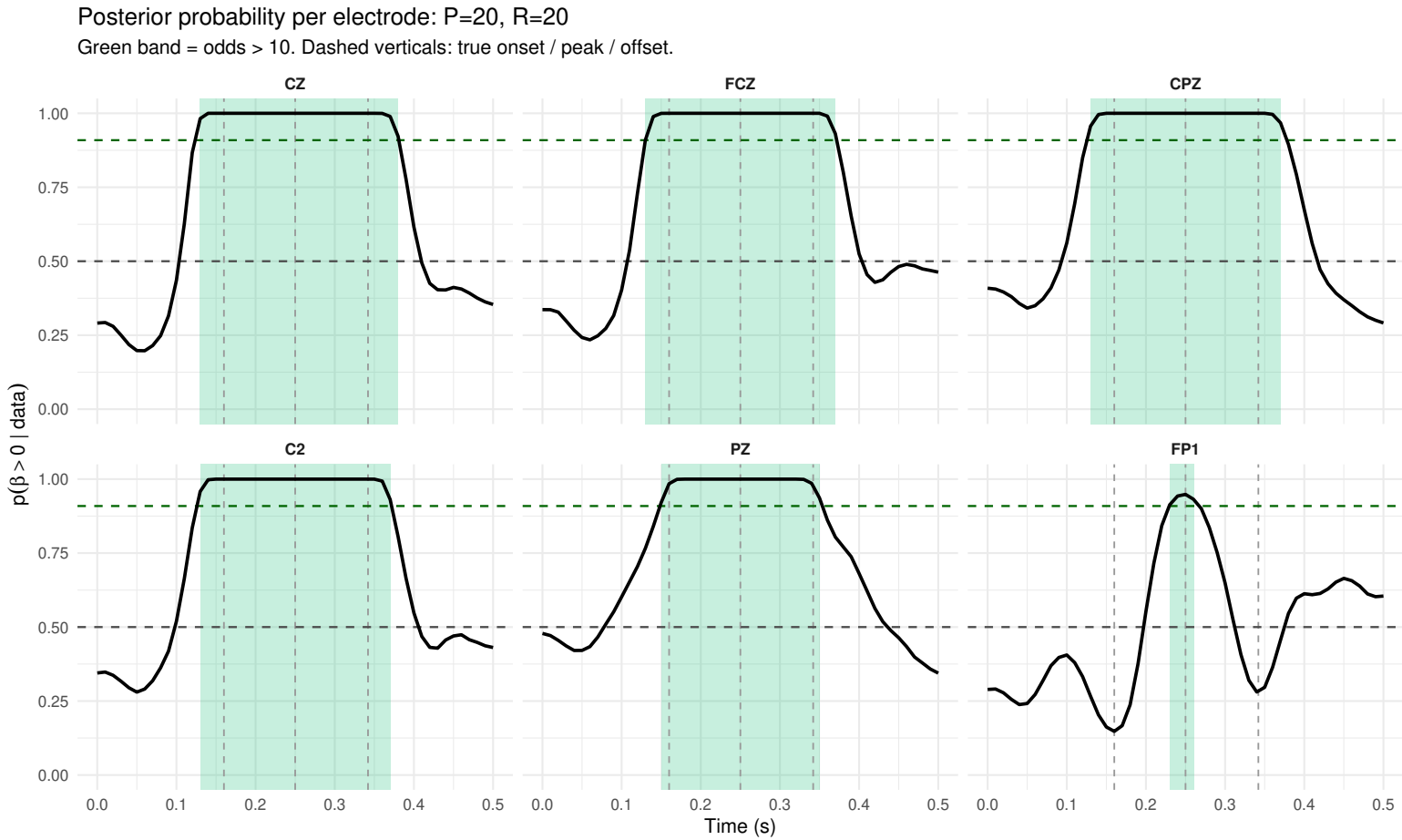
Modeling

```
bgam_fit <- brm(
    formula = eeg_mean | se(eeg_sd) ~
        condition +
        (1 + condition | participant) +
        t2(xproj, yproj, time,
           by   = condition,
           bs   = c("tp", "tp"),
           k    = c(20, 20),
           d    = c(2, 1),
           full = FALSE),
    data    = sim_data_model,
    family  = gaussian(),
    chains  = 4,
    cores   = 4,
    iter    = 3000,
    warmup  = 1000,
    seed    = 42,
    threads = threading(threads = 4),
    backend = "cmdstanr",
    control = list(adapt_delta = 0.99, max_treedepth = 13)
)
```

How the model fits the data ($P = 20,\ k = (20, 20)$)



Posterior probability per electrode ($P = 20,\ k = (20, 20)$)



Model result: $P = 20$, $R = 20$, $k = (20, 20)$

Fit: `bgam_fit_v5_k20_n20_r20_sd.rds`

Diagnostic target	Result	Interpretation
Convergence	PASS	0 divergences; $\widehat{R}_{\max} = 1.003$
Onset / offset	150–350 ms	truth 160–342 ms; bias $-10/+8$ ms
Spatial top channel	CZ	correct seed electrode
Spatial F1	0.847	25 TP, 9 FP, 0 FN
Spatiotemporal F1	0.831	343 TP, 8 FP, 132 FN
Peak recovery	76.9%	pointwise peak is underestimated
Window-mean recovery	105.6%	95% CrI covers truth
Marginal condition diff.	0.0529	empirical diff. = 0.0533
Wall clock	173 h	total execution time on LIDO HPC

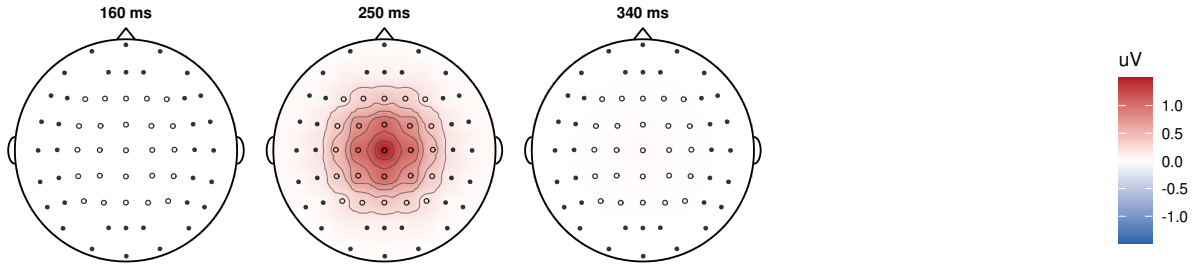
This result covers the temporal window, spatial localisation, and window-averaged amplitude. The pointwise peak is still underestimated.

Topomaps over the signal window ($P = 20$, $R = 20$)

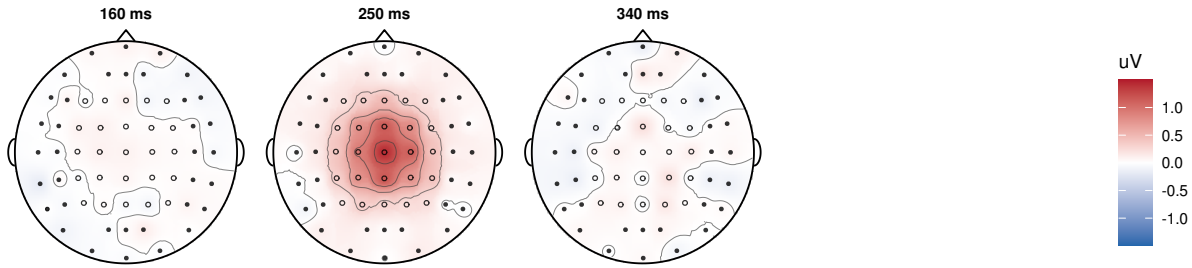
Topomaps over the signal window: P=20, R=20

Columns: onset / peak / offset. Rows 1-2 show cond2 - cond1; residual = empirical - model.

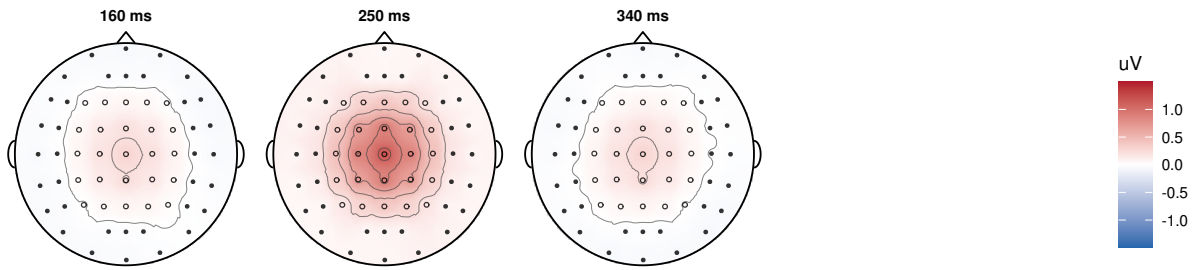
0. Ground truth



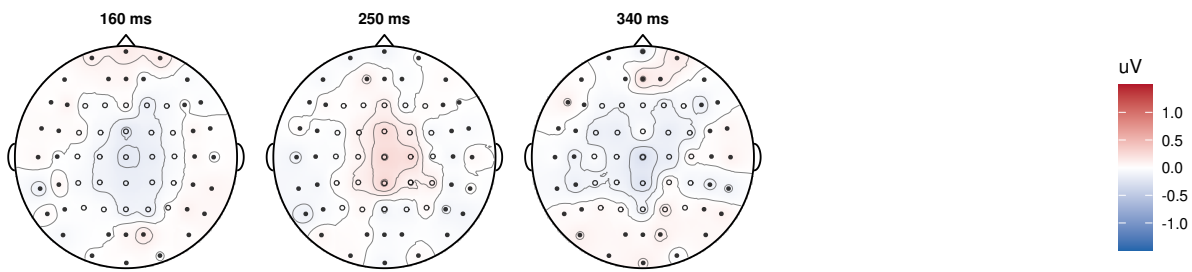
1. Empirical difference



2. BGAM estimate



3. Residual



SE calibration: effective trial count

Data/model: $P = 5$ participants, $R = 20$ trials per condition, 61 electrodes, 51 time points, $k = (20, 20)$.

Definitions for one participant $\times$ channel $\times$ condition $\times$ time cell:

$$\mathrm{SD} = \text{trial-level variability}, \qquad \mathrm{SEM} = \frac{\mathrm{SD}}{\sqrt{R}}, \qquad \mathrm{SE}_{\mathrm{eff}} = \frac{\mathrm{SD}}{\sqrt{R_{\mathrm{eff}}}}.$$

$R_{\mathrm{eff}} \leq R$ is the effective number of independent trials after trial dependence.

Here: $R_{\mathrm{eff}} = 14$ comes from an empirical trial-correlation ESS check in the active window/active channels ($13.77 \approx 14$). $R_{\mathrm{eff}} = 10$ is a more conservative intermediate sensitivity value.

Fit	R_{eff}	Onset bias	Offset bias	Peak rec.	Mean rec.	Spatial F1	ST F1	LOOic	Wall clock
`k20_reff10`	10	0 ms	−12 ms	86.4%	105.7%	0.806	0.814	1289.4	69 h
`k20_reff14`	14	0 ms	−12 ms	88.1%	105.9%	0.794	0.816	−2311.5	68 h

Both fits recover the temporal window and CZ-centred spatial effect. Peak is still underestimated, but window-mean amplitude is close to truth.

Methods comparison ($P = 5$ pilot, 30 MC)

Following Nalborczyk & Bürkner (2025) Table 1. All rows are 30-MC summaries from the $P = 5$ pilot setting. Onset/offset in seconds; SD across 30 sims.

Onset estimates (truth = 0.160 s):

Method	Bias (s)	MAE (s)	RMSE (s)	SD (s)	$F1_{ST}$ (mean ± SD)
Raw p	+0.010	**0.010**	0.012	0.008	0.462 ± 0.014
BGAM_2D	−0.011	0.011	**0.012**	**0.005**	**0.799 ± 0.007**
TFCE	+0.013	0.014	0.017	0.011	0.539 ± 0.105
Cluster	+0.016	0.016	0.017	0.007	0.514 ± 0.020
FDR (BH)	+0.063	0.063	0.065	0.016	n/a
Holm	+0.093	0.093	0.094	0.012	n/a

Offset estimates (truth = 0.342 s):

Method	Bias (s)	MAE (s)	RMSE (s)	SD (s)
BGAM_2D	+0.012	**0.012**	**0.014**	**0.006**
Raw p	−0.006	0.013	0.015	0.014
TFCE	−0.015	0.015	0.018	0.011
Cluster	−0.018	0.019	0.020	0.009
FDR (BH)	−0.068	0.068	0.070	0.017
Holm	−0.082	0.082	0.082	0.010

FP and FN definitions

This checks the previous onset/offset summary metrics using the full time-by-channel detection mask.

Each cell is one time-by-channel location (t, c).
The ground-truth mask is:

$$M_{t,c} = \begin{cases} 1, & t \in [0.160, 0.342] \text{ s and } c \text{ is an active electrode,} \\ 0, & \text{otherwise.} \end{cases}$$

A detected cell is a cell declared significant by a given method. For detection, we use $p_{t,c} < \alpha$, where $\alpha = 0.10$.

Let $D_{t,c} = 1$ if cell (t, c) is detected, and $D_{t,c} = 0$ otherwise.

$$\begin{aligned} TP &: \ D_{t,c} = 1, \ M_{t,c} = 1, \quad & FP &: \ D_{t,c} = 1, \ M_{t,c} = 0, \\ FN &: \ D_{t,c} = 0, \ M_{t,c} = 1, \quad & TN &: \ D_{t,c} = 0, \ M_{t,c} = 0. \end{aligned}$$

$$\text{FP rate} = \frac{FP}{FP + TN}, \qquad \text{FN rate} = \frac{FN}{FN + TP}.$$

FP/FN diagnosis ($\alpha = 0.10$)

Onset/offset metrics only summarise the first and last detected time points. They do not show whether the full detection mask contains many false-positive cells or misses most true cells.

Method	FP rate	FN rate	$F1_{ST}$	Mean FP cells	FP-only clusters
Raw p	0.102	0.530	0.462	268.9	141.4
BH	0.001	0.943	0.106	3.0	2.9
Holm	0.000	0.996	0.008	0.2	0.2
Cluster	0.003	0.648	0.514	8.0	0.0
TFCE	0.007	0.608	0.539	18.4	1.1

Raw p has the highest FP rate because each time-channel cell is tested separately without correction, creating many scattered detections (141.4 FP-only clusters; 113.0 outside-window clusters). BH and Holm strongly control false positives, but become too conservative and miss most true cells. Cluster and TFCE use spatio-temporal structure, so they reduce scattered FP cells and give a better FP/FN trade-off.

Appendix: Metric definitions

$F1$ measures detection accuracy of active cells.

$$F1 = \frac{2\,P\,R}{P+R}, \quad P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}.$$

P = fraction of predicted-positive cells that are truly active.
R = fraction of truly active cells that are predicted positive.

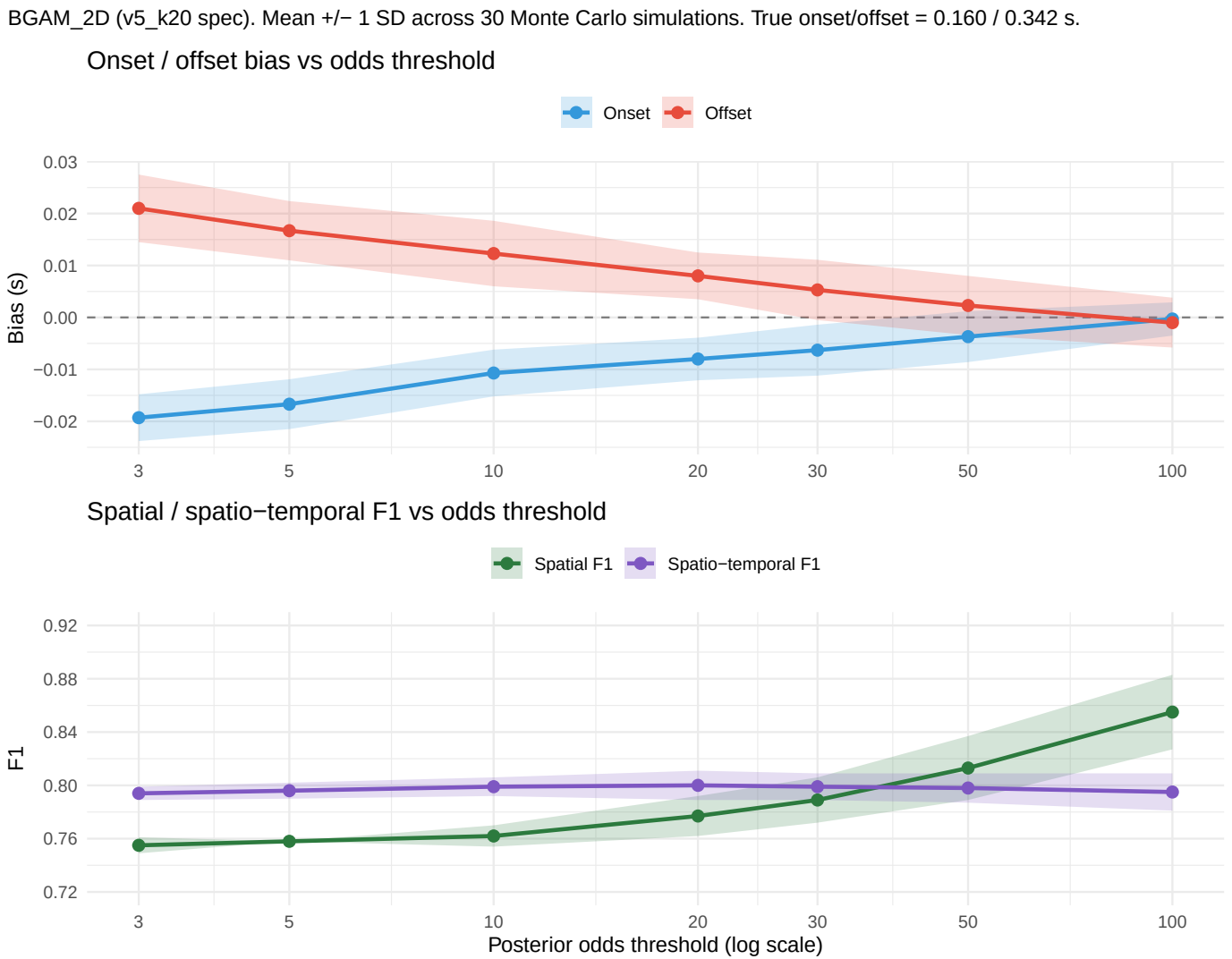
A cell is **predicted positive** iff its posterior odds $> \theta$. This is the threshold the BGAM uses to declare a cell "significant", and it defines the TP, FP, FN counts above.

The two metrics differ only in what counts as a "cell":

	Spatial F1	**Spatio-temporal F1**
Cell	channel c	(channel, time) pair (c, t)
Evaluated at	peak time $t^\star = 0.251\,\text{s}$	all 51 time points
Total cells	61 channels	$61 \times 51 = 3{,}111$
True positives	25 active channels	$25 \times 19 = 475$ active cells

Appendix: BGAM_2D odds sensitivity

Setting: $P = 5$ pilot, 30 Monte Carlo simulations, $v5_k20$ specification.



Appendix: BGAM_2D posterior-mask diagnosis

Setting: $P = 5$ pilot, 30 Monte Carlo simulations.

We define the following rule to obtain the BGAM_2D posterior mask:

$$\text{posterior odds}_{t,c} = \frac{P(\Delta_{t,c} > 0 \mid y)}{P(\Delta_{t,c} \leq 0 \mid y)} > 10.$$

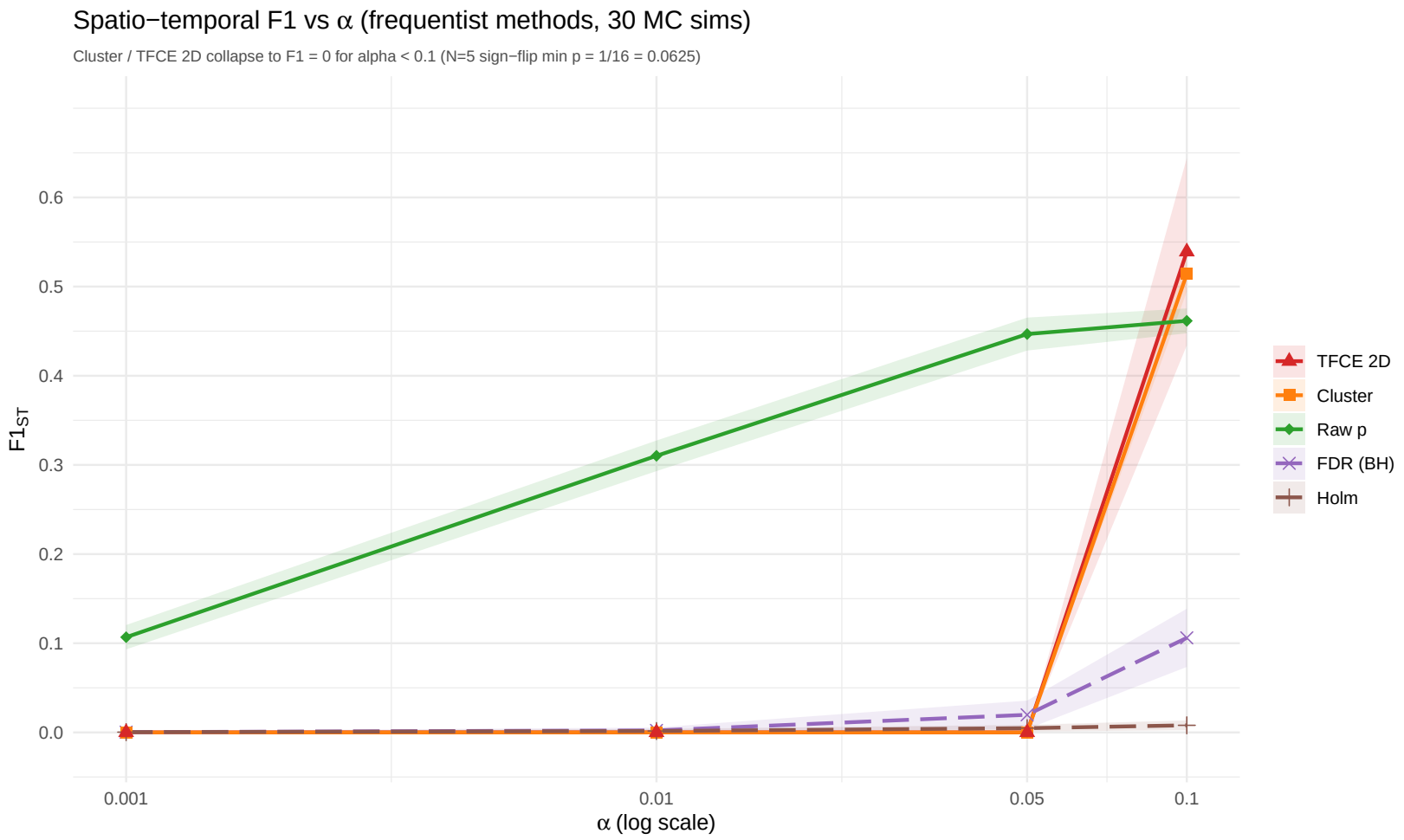
$$\Delta_{t,c} = \text{condition difference at cell } (t, c) \quad (\text{cond2} - \text{cond1}).$$

Method	Odds thr.	Mean TP	Mean FP	Mean FN	FP rate	FN rate	Precision	$F1_{ST}$
BGAM_2D (se=SD)	10	390.1	111.3	84.9	0.042	0.179	0.778	0.799

BGAM_2D has a higher FP rate than cluster and TFCE, but a much lower FN rate. This means it recovers most of the true spatio-temporal effect, while also extending somewhat beyond the strict ground-truth mask.

This pattern is expected from the smooth Bayesian surface: information is shared across neighbouring times and channels, which reduces missed true cells, but can create smooth spillover around the true active region.

Appendix: α sensitivity (frequentist methods)



Appendix: Larger P improves data-to-parameter ratio

Spline coefficients (v5_k20): $\underbrace{20}_{\text{tp loc.}} \times \underbrace{20}_{\text{tp time}} \times \underbrace{2}_{\text{cond.}} = 800.$

Signal-bearing observations: $\underbrace{20}_{\text{ppt}} \times \underbrace{25}_{\text{active}} \times \underbrace{19}_{\substack{\text{signal}\\\text{steps}}} \times \underbrace{1}_{\text{cond2}} = 9{,}500.$

Ratio: $9{,}500/800 \approx 12$, now within the lower end of the $\geq$ 10–20 obs/param rule of thumb (Peduzzi 1996; Wood 2017).

With $P = 20$, the likelihood carries more information for the same spatio-temporal smooth. This helps recover the temporal window and spatial extent, while the smoothness penalty can still attenuate sharp pointwise peaks.

Remaining caveat: peak recovery is still affected by smoothing bias.

Appendix: Wand–Jones smoothing bias

Any spline averages each point with its neighbours over a bandwidth h. Near a peak the neighbours are lower than the peak, so the smoothed estimate sits *below* the truth. Plugging in our setting ($h \approx 0.025\,\mathrm{s}$, $A = 1.5\,\mu\mathrm{V}$, $\sigma_t = 0.060\,\mathrm{s}$) gives a peak under-estimate of ~ 0.1–$0.3\,\mu\mathrm{V}$ $\Rightarrow$ a **peak-recovery ceiling of** ~ 85–95%, even for an ideal BGAM.

Formal statement (Wand & Jones 1995, Ch. 4):

$$E\big[\hat{f}(x_0)\big] - f(x_0) \;\approx\; \tfrac{1}{2}\, h^2\, f''(x_0)\, \mu_2(K).$$

h = effective bandwidth (here $\approx T/k_t = 0.5/20 = 0.025\,\mathrm{s}$); at the peak $f'' < 0 \Rightarrow$ negative bias.

This hits *Peak recovery* (ceiling ~ 85–95%) but is mostly averaged out in *Mean over window* (ceiling ~ 95–98%), because f'' flips sign between centre and edges.

Appendix: pilot simulated EEG data ($P = 5$, se(eeg_sd))



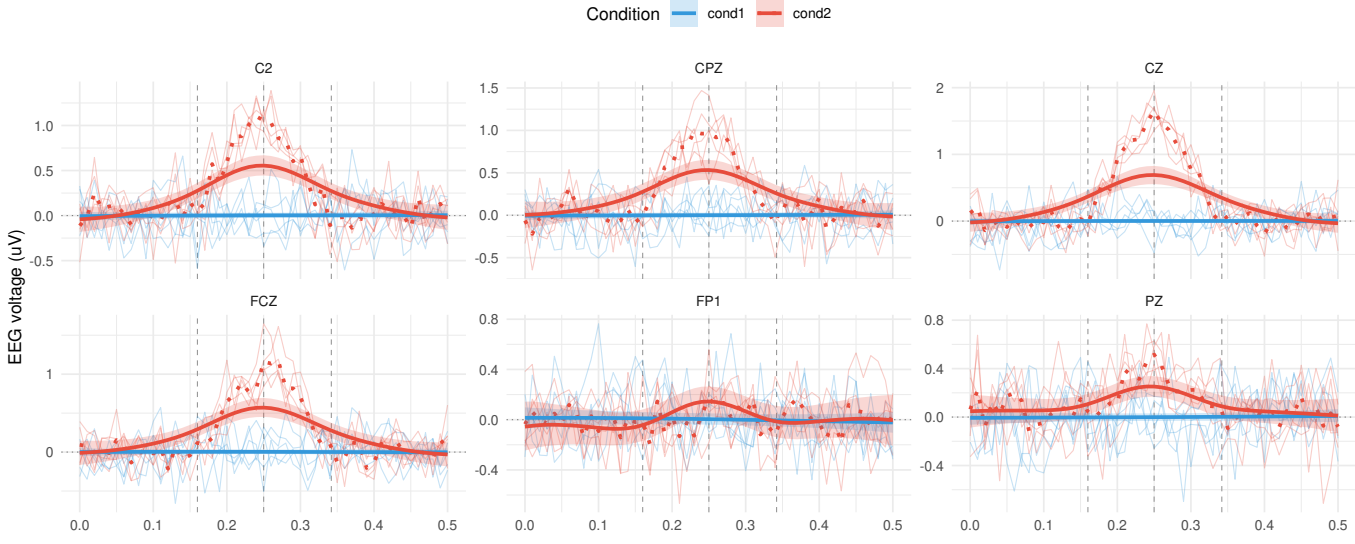
Appendix: pilot predictions vs raw ($P = 5$, se(eeg_sd))

Predictions vs raw — bgam_fit_v5_k20

k = c(20, 20), full = FALSE

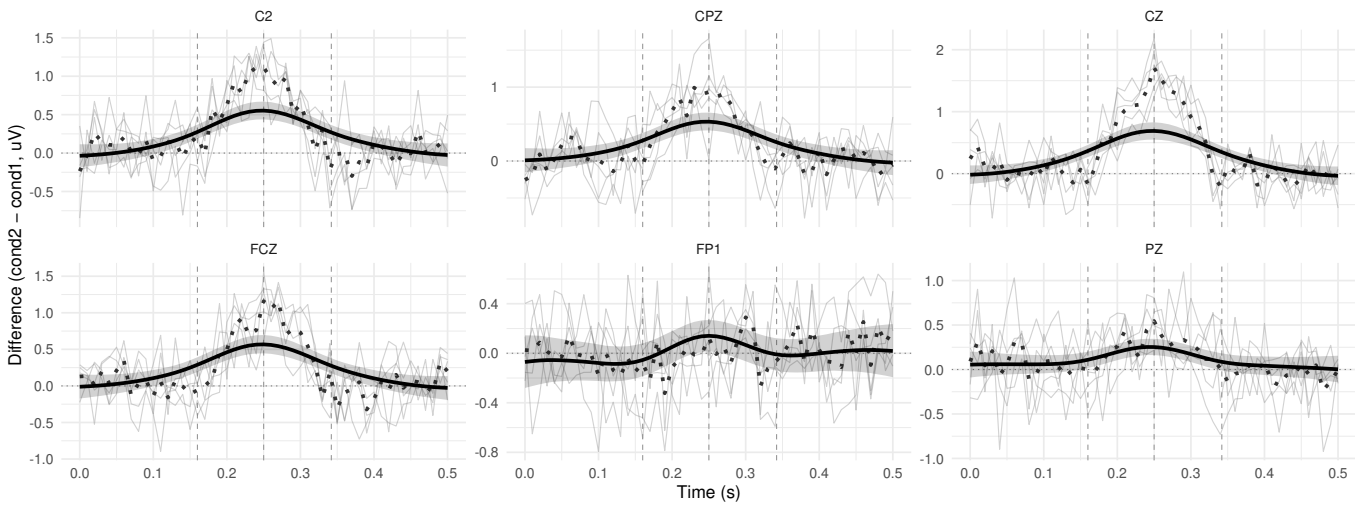
Section 1 — raw vs model predictions, per condition

Thin colored lines: per-ppt. Dotted (cond2 only): raw grand mean. Solid + ribbon: model: posterior mean + 95% CrI. Dashed verticals: 0.160 / 0.250 / 0.342 s.

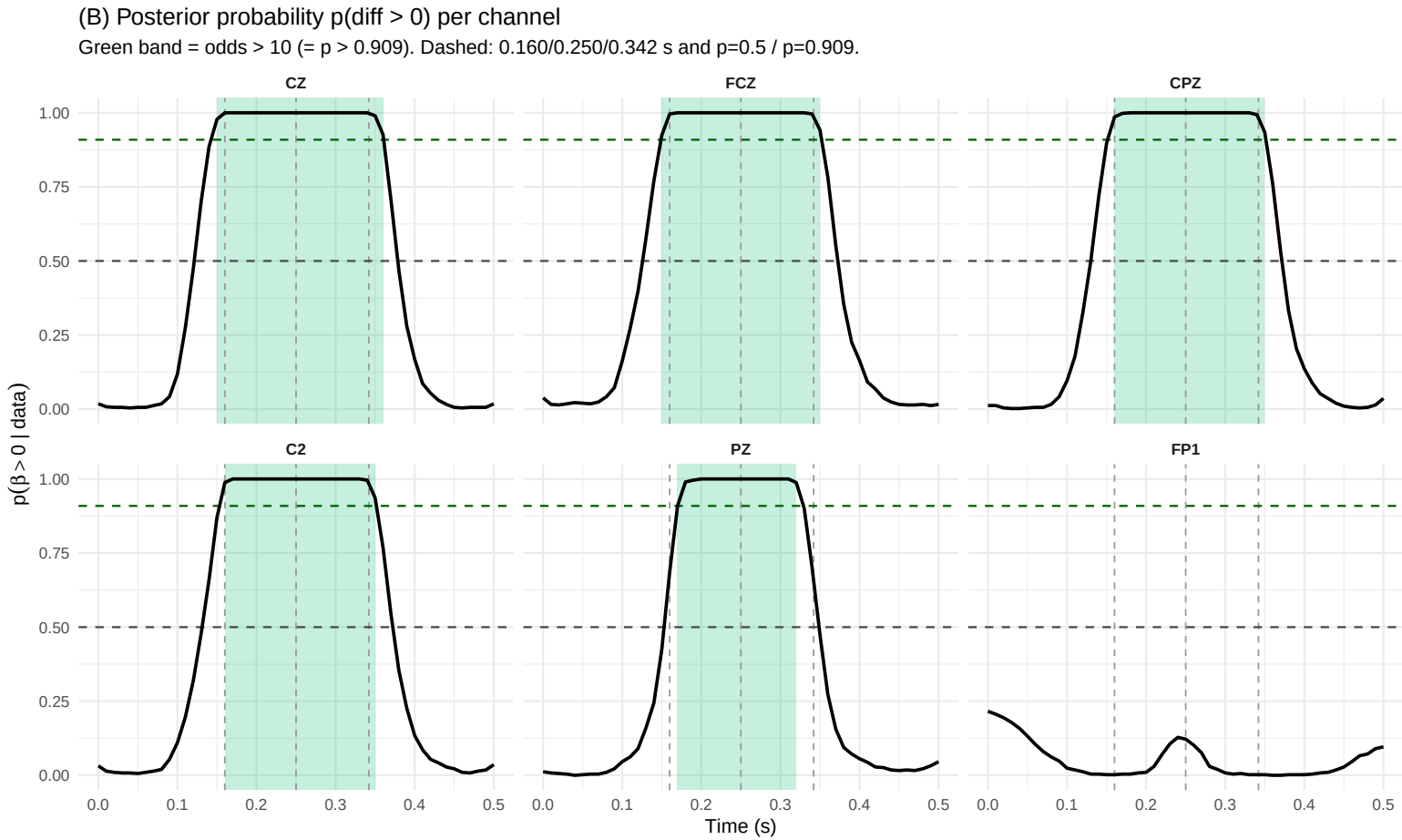


Section 2 — condition difference (cond2 – cond1)

Thin grey: per-ppt. Dotted: raw grand mean diff. Solid black + ribbon: posterior mean + 95% CrI.



Appendix: pilot posterior probability ($P = 5$)



Appendix: pilot model comparison ($P = 5$)

Original pilot sweep over basis dimensions and model structure. Here, **pilot** means the early smaller setting with $P = 5$ participants and $R = 20$ trials per condition.

k_{xy}	k_t	`full`	SE	Onset bias (s)	Offset bias (s)	Peak rec. (%)	Mean rec. (%)	Spatial F1	LOOic	Wall-clock
12	12	F	SD	−0.070	+0.068	46	55.4	0.714	57462.5	5 h
20	20	F	SD	−0.070	+0.068	46	54.9	0.714	57465.5	14 h
20	**20**	**F***	SD	**−0.080**	**+0.078**	40	**75.5**	**0.758**	**57462.4**	28 h
20	**20**	**F**	**SEM**	**+0.010**	**−0.012**	**89**	**106.1**	**0.962**	–	67 h
10	30	T	SD	−0.080	+0.068	36	54.3	0.714	57469.4	48 h
20	30	T	SD	−0.130	+0.108	40	63.2	0.658	57468.7	112 h
30	20	T	SD	−0.150	+0.138	58	57.7	0.633	57490.4	69 h
30	30	T	SD	−0.150	+0.138	36	57.7	0.625	57489.9	110 h

* Winner among SD fits in the original pilot comparison.

Appendix: spatial ground-truth alternative

A less arbitrary alternative is to define the active region by the half-maximum level set, following the FWHM convention for Gaussian peaks ([1]):

$$\text{electrode } c \text{ is active if } w(c) \geq 0.5\, w_{\max}, \qquad w_{\max} = \max_{\text{all electrodes } e} w(e).$$

This keeps electrodes within the half-maximum spatial support and is invariant to changes in the maximum signal amplitude.

For EEG electrodes, the final spatial ground truth can combine two parts:

- **Half-maximum amplitude rule**: define the core active region from the simulated Gaussian surface.
- **Electrode-layout correction**: adjust the grid-based region using prior knowledge about the electrode layout and expected spatial spread.

[1] Weisstein, Eric W. "Gaussian Function." From *MathWorld*–A Wolfram Resource. https://mathworld.wolfram.com/GaussianFunction.html. The FWHM section derives the half-maximum points of a Gaussian and defines the width between them.

Appendix: Onset / offset bias

At seed electrode CZ, find times where condition difference is "significantly positive" (posterior odds > 10, Jeffreys), then take the temporal extent.

```
epred  = add_epred_draws(grid_at_CZ, fit, re_formula = NA)
diff   = cond2 - cond1                          # per draw, per time
p_sig  = mean(diff > 0)                         # per time
odds   = p_sig / (1 - p_sig)

onset  = min{ t : odds(t) > 10 }
offset = max{ t : odds(t) > 10 }

bias_onset  = onset  - 0.160 s
bias_offset = offset - 0.342 s
```

Truth: $(0.160, 0.342)$ s (boundary of truncated Gaussian $g(z)$ where $|z| = 1.5$).

Appendix: Peak & Mean recovery (at seed CZ)

Peak recovery (Target 4) — point estimate at peak time.

```
peak_t   = (onset + offset) / 2 = 0.251 s
diff_t   = posterior(cond2 - cond1) at (CZ, peak_t)
est_peak = mean(diff_t);  CI = quantile(diff_t, c(.025, .975))
rec_pct  = 100 * est_peak / 1.5     # truth = 1.5 uV (g(0)=1)
```

Mean recovery (Target 4b) — window-mean.

```
window   = [0.160, 0.342] s
# per-draw mean across the window (then aggregate across draws)
mean_d   = per_draw: mean_t( diff(t in window) )
est_mean = mean(mean_d);  CI = quantile(mean_d, c(.025, .975))
rec_pct  = 100 * est_mean / 0.73    # truth = window-mean of g(z)
```

Appendix: LOOic

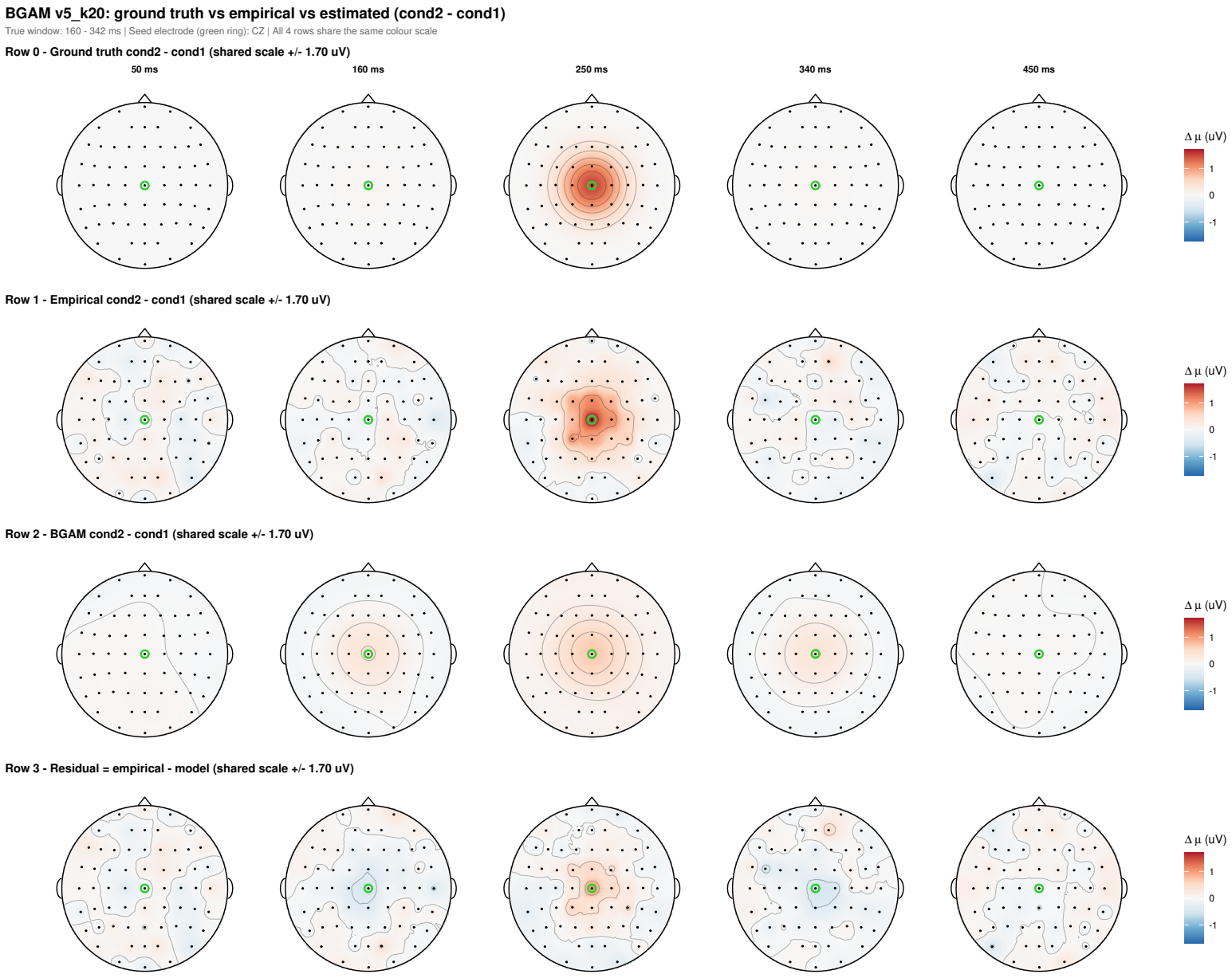
Bayesian out-of-sample predictive accuracy via PSIS-LOO (Vehtari, Gelman, Gabry 2017).

```
lo  = loo(fit, moment_match = FALSE)

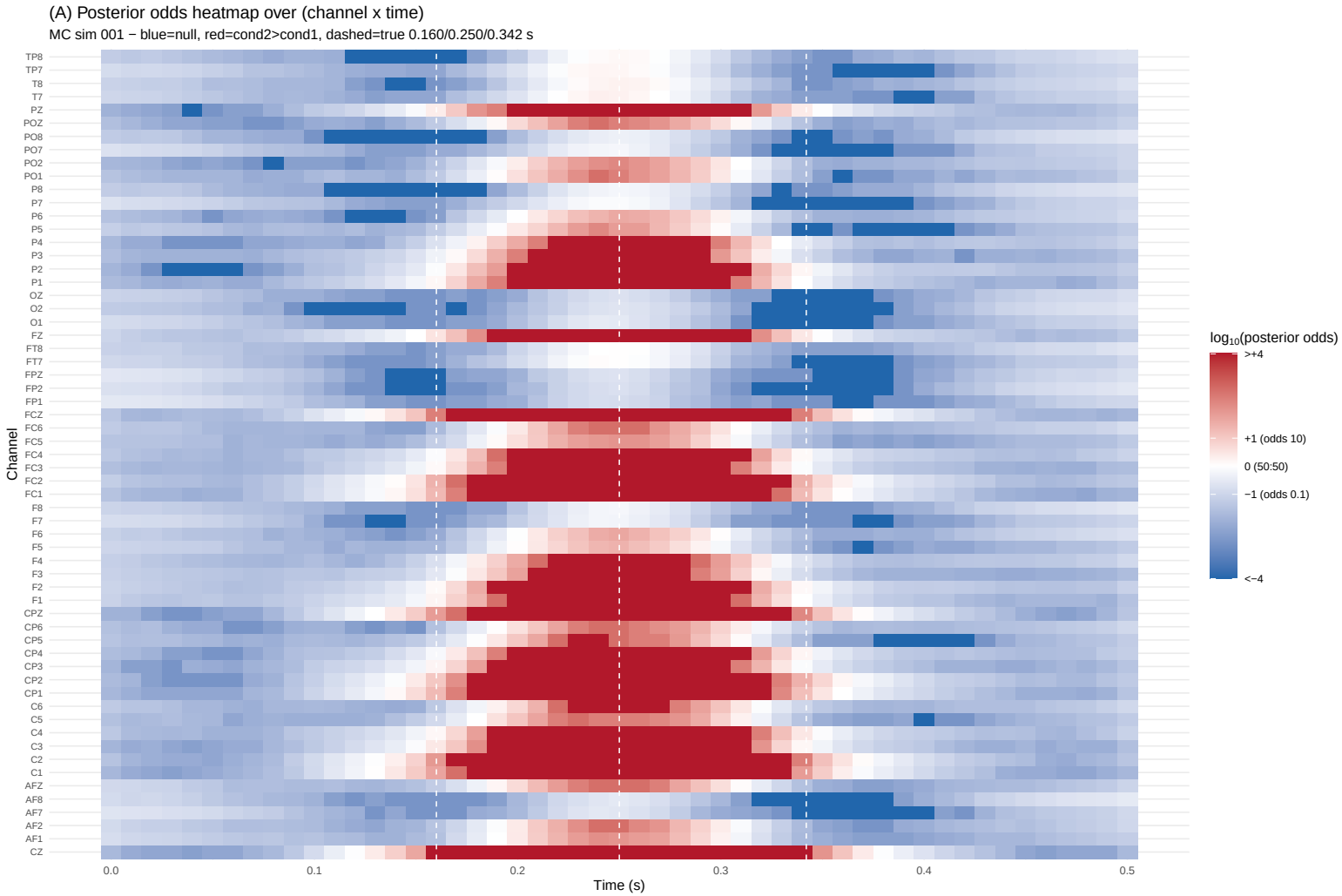
# lo$pointwise[i, "elpd_loo"]  for i = 1 .. 31,110 cells
# elpd_loo_i = log p(y_i | y_{-i})  estimated by PSIS

elpd_loo = sum(lo$pointwise[, "elpd_loo"])
looic    = -2 * elpd_loo
```

Appendix: Topomap (ground truth vs empirical vs model; $P = 5$, se(eeg_sd))



Appendix: Posterior probability ($k = (20, 20)$, `full=F`; $P = 5$, se(eeg_sd))



Appendix: Posterior $\log_{10}$ odds per electrode ($P = 5$, se(eeg_sd))

